In this paper, we denote that $a_m : \mathbb{N} \to \mathbb{R}$, $(m \in \mathbb{N})$.

Def. A set of sequences that converges to zero

$$C_0(\mathbb{N}) \stackrel{\text{def}}{=} \{ a : \mathbb{N} \to \mathbb{R} \mid \lim_{n\to\infty} a(n) = 0 \}$$

And define the ℓ^1-space

$$\ell^1(\mathbb{N}) \stackrel{\text{def}}{=} \left\{ a \in C_0(\mathbb{N}) \;\middle|\; \text{the series } \sum_{n=1}^{\infty} a(n) \text{ converges absolutely} \right\},$$

It is a normed space, with a norm

$$\|a\|_1 = \sum_{n=1}^{\infty} |a(n)|.$$

Thm. The normed space $\ell^1(\mathbb{N})$ is complete metric space.

Proof. Let $\{a_m : \mathbb{N} \to \mathbb{R}\}$ be a Cauchy sequence. Then, for each $n \in \mathbb{N}$,

$$|a_m(n) - a_k(n)| \leq \sum_{n=1}^{\infty} |a_m(n) - a_k(n)| = \|a_m - a_k\|_1 \quad \text{for any } m, k \in \mathbb{N}.$$

So, since $\{a_m\}$ is Cauchy sequence, $\{a_m(n)\}$ is also Cauchy sequence for each $n \in \mathbb{N}$.
The $\mathbb{R}$ is complete, define $a(n) = \lim_{m\to\infty} a_m(n)$ for each $n \in \mathbb{N}$.
To show that $a \in \ell^1(\mathbb{N})$ and $\|a_m - a\|_1 \to 0$ as $m \to \infty$, let $\varepsilon > 0$ be given.
Since $\{a_m\}$ is Cauchy sequence, there exists $N \in \mathbb{N}$ such that

$$m, k \geq N \Rightarrow \|a_m - a_k\|_1 < \varepsilon.$$

Then, for each $n \in \mathbb{N}$,

$$m, k \geq N \Rightarrow \sum_{i=1}^{n} |a_m(i) - a_k(i)| \leq \|a_m - a_k\|_1 < \varepsilon.$$

Letting $k \to \infty$, we obtain that, for all $n \in \mathbb{N}$

$$m \geq N \Rightarrow \sum_{i=1}^{n} |a_m(i) - a(i)| \leq \varepsilon.,$$

So $\|a_m - a\|_1 \to 0$ as $m \to \infty$.

In particular, for each $n \in \mathbb{N}$,

$$\sum_{i=1}^{n} |a(i)| \leq \sum_{i=1}^{n} |a_N(i)| + \sum_{i=1}^{n} |a(i) - a_N(i)| \leq \|a_N\|_1 + \varepsilon, \text{ so } a \in \ell^1(\mathbb{N}).$$

$\boxtimes$